

Subsampling Confidence Bound for Persistent Diagram via Time-delay Embedding

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The Periodicity Detection Problem

- Natural and artificial systems inherently produce recurring patterns (e.g., biological rhythms, climate cycles).
- Traditional methods
 - ▶ Fourier transform or Spectral analysis based methods.
 - ▶ Testing based on sliding window embeddings, **SW1pers algorithm**
- Limitations
 - ▶ Lack of statistical quantification.
 - ▶ Lack of a rigorous understanding of how periodicity affects topology of sliding windows.

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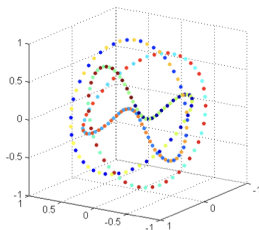
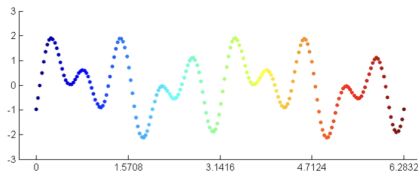
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Time delay Embedding

- Given a continuous function $f \in C(\mathbb{T}, \mathbb{R})$, for an embedding dimension m and window size τ ,

$$SW_{m,\tau}f(t) = \begin{bmatrix} f(t) \\ f(t + \tau) \\ \vdots \\ f(t + m\tau) \end{bmatrix} \in \mathbb{R}^{m+1}$$

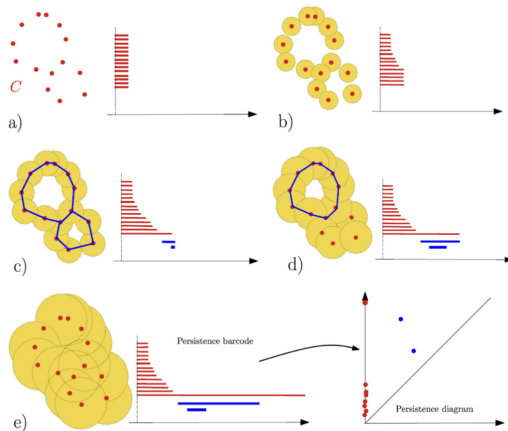
- For a purely periodic signal and appropriately chosen m, τ , the point cloud traces a loop that is topologically equivalent to S^1 .



Perea & Harer, "Sliding Windows and Persistence," Found. Comput. Math., 2015.

Persistence Diagrams

- Persistent diagram captures topological features via filtration.



Chazal & Michel, "An Introduction to TDA," Front. Artif. Intell., 2021.

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Periodicity

- True sampling function $f : \mathbb{R} \rightarrow \mathbb{R}$.
- Ξ - periodic: If for any $t \in \mathbb{R}$,

$$f(t + \Xi) = f(t).$$

- Non-periodic: If for any $t_1 \neq t_2 \in \mathbb{R}$,

$$f(t_1) \neq f(t_2) \text{ or } f'(t_1) \neq f'(t_2).$$

- Periodic functions and Non-periodic functions can behave very similarly!

Periodicity

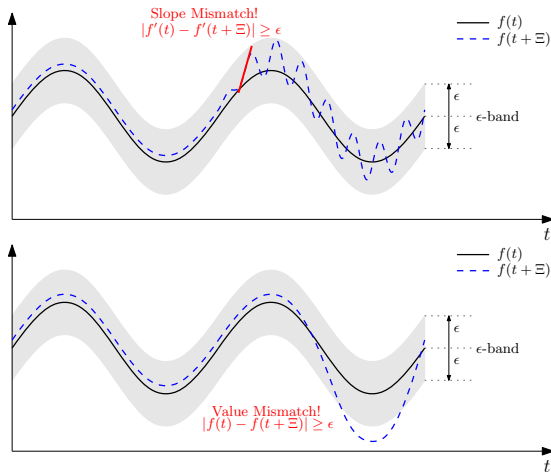
- Introduce ϵ -distinguishability.
- f is (Ξ, ϵ) -periodic if f is Ξ -periodic, and there exists $K > 0$ such that for any $t_1, t_2 \in \mathbb{R}$ with $\min_{n \in \mathbb{Z}} |t_1 - t_2 + n\Xi| \geq K\epsilon$,

$$|f(t_1) - f(t_2)| \geq \epsilon \text{ or } |f'(t_1) - f'(t_2)| \geq \epsilon.$$

- f is ϵ -non-periodic if there exists some $K > 0$ such that for all $t_1, t_2 \in \mathbb{R}$ with $|t_1 - t_2| \geq K\epsilon$,

$$|f(t_1) - f(t_2)| \geq \epsilon \text{ or } |f'(t_1) - f'(t_2)| \geq \epsilon.$$

Periodicity



Park, An, Kim, & Kim, "Subsampling Confidence Bound for PD via Time-delay Embedding," 2026.

Statistical Model and Assumptions

- Points are sampled i.i.d. from a uniform distribution on $\mathbb{T} = [T_{min}, T_{max}]$, we can observe $\{SW_{m,\tau}f(t_j)\}$
- Assumption 1. The true sampling function is either (Ξ, ϵ) -periodic for some $\Xi, \epsilon > 0$, or ϵ -non-periodic for some $\epsilon > 0$.
- Assumption 2. there exists some $\delta > 0$ such that for all $t \in \mathbb{R}$, either $|f'(t)| \geq \delta$ or $|f''(t)| \geq \delta$, and for all $t \in \mathbb{R}$, $|f'(t)| \leq L_1$ $|f''(t)| \leq L_2$.
- Assumption 3. For all t_1, t_2 with $|t_1 - t_2| \leq K\epsilon$, $|f(t_1) - f(t_2)| \geq \delta|t_1 - t_2|$ or $|f'(t_1) - f'(t_2)| \geq \delta|t_1 - t_2|$ holds.

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Topology result for non-periodic function

Theorem

Suppose $f \in C^2$ satisfies assumptions and ϵ -non-periodic. Then there exists some constant $C_{\delta,L}$ depending only on δ and L such that as $\tau \rightarrow 0$, $SW_{m,\tau}f$ satisfies that for any

$$0 < t < \frac{\sqrt{m+1}}{2} \min \{\epsilon, C_{\delta,L}\}$$

we have that $(SW_{m,\tau}f(\mathbb{T}))^t$ is homotopic to interval, and in particular contractible.

Corollary

Let $\mathcal{P}_1(SW_{m,\tau}f)$ be 1-dimensional persistent homology of $SW_{m,\tau}f$ of Čech complex filtration, understood as a subset of $\mathbb{R}^2$. Then

$$\mathcal{P}_1(SW_{m,\tau}f) \cap [0, a) \times \mathbb{R} = \emptyset.$$

Topology result for periodic function

Theorem

Let $\Xi > 0$, and Suppose $f \in C^2$ satisfies assumptions and (Ξ, ϵ) -periodic. Then $SW_{m,\tau}f$ satisfies that for any

$$0 < t < \frac{\sqrt{m+1}}{2} \min \{\epsilon, C_{\delta,L}\}$$

we have that $(SW_{m,\tau}f(\mathbb{T}))^t$ is homotopic to a circle S^1 .

Corollary

Let $\mathcal{P}_1(SW_{m,\tau}f)$ be 1-dimensional persistent homology of $SW_{m,\tau}f$ of Čech complex filtration, understood as a subset of $\mathbb{R}^2$. Then

$$\mathcal{P}_1(SW_{m,\tau}f) \cap [0, a) \times \mathbb{R} = \{(0, d)\},$$

$$d \geq a = \frac{\sqrt{m+1}}{2} \min \{\epsilon, C_{\delta,L}\}.$$

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Subsampling Confidence Bound

- Apply existing subsampling based confidence bound method on manifold with positive reach.
- For b possible subamples, define

$$L_{m,\tau}(t) = \frac{1}{\binom{n}{b}} \sum_{j=1}^{\binom{n}{b}} I\left(d_H(X_{m,\tau}, X_{m,\tau,b}^{(j)}) > t\right)$$

- Fasy et al. 2014, let $b = o(n/\log n)$ and $c_\alpha = 2(L_{m,\tau})^{-1}(\alpha)$,

$$P\left(d_B(\mathcal{P}(X_{m,\tau}), \mathcal{P}(M_{m,\tau})) > c_\alpha\right) \leq \alpha + O\left(\frac{b}{n}\right)^{1/4}.$$

Confidence bound analysis for embedding dimension

- Suppose f is (Ξ, ϵ) periodic function. Under the relation $(m+1)\tau = \Xi$, additional structure analysis are possible.

Theorem

For fixed n, b , $\tilde{c}_\alpha^m = \frac{1}{\sqrt{m+1}} c_\alpha$ converges as $m \rightarrow \infty$.

Theorem

(a) The sequences of persistence diagrams $\mathcal{P}(\tilde{X}_{m,\tau_m})$ and $\mathcal{P}(\tilde{M}_{m,\tau_m})$ form Cauchy sequences with respect to the bottleneck distance.

(b) As $b = o(n/\log n)$ tending to infinity as $n \rightarrow \infty$, $\tilde{c}_\alpha = \lim_{m \rightarrow \infty} \tilde{c}_\alpha^m$ satisfies

$$P\left(d_B(\mathcal{P}_\infty(\tilde{X}), \mathcal{P}_\infty(\tilde{M})) > \tilde{c}_\alpha\right) \leq \alpha + O\left(\frac{b}{n}\right)^{1/4}$$

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Hypothesis Testing

- Binary hypothesis test of periodicity.
 - ▶ $H_0 : f$ is (Ξ, ϵ) -non-periodic function.
 - ▶ $H_1 : f$ is (Ξ, ϵ) -periodic function.

- Rejection region in the persistence diagram plane

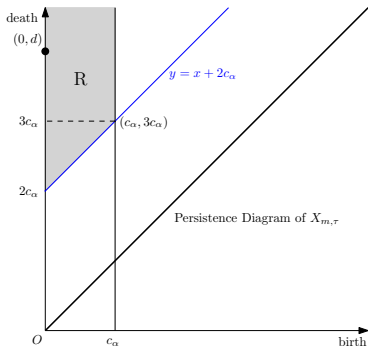
$$R := \{(b, d) \in \mathcal{P}(X_m) \mid (b < c_\alpha) \text{ and } (d - b \geq 2c_\alpha)\}.$$

Hypothesis Testing

$$R := \{(b, d) \in \mathcal{P}(X_m) \mid (b < c_\alpha) \text{ and } (d - b \geq 2c_\alpha)\}.$$

- Decision function

$$\phi(X_m) = \begin{cases} 0 & \text{if } |\mathcal{P}(X_m) \cap R| = 0 \quad (\text{Accept } H_0) \\ 1 & \text{if } |\mathcal{P}(X_m) \cap R| \geq 1 \quad (\text{Reject } H_0) \end{cases}$$



Type I error control

Theorem

For given $\alpha \in (0, 1)$, the test has asymptotic significance level α , i.e.,

$$P(\phi(X_m) = 0) \leq \alpha.$$

Theorem

For given $\alpha \in (0, 1)$, the test has type II error asymptotically bounded by α , i.e.,

$$P(\phi(X_m) = 1) \leq \alpha.$$

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Synthetic Data

- Four sampling functions with four types of error

$$f_1(x) = \frac{3}{2 - \cos(5x)}$$

$$f_2(x) = 5 \log(\sin(10x) + 5) + e^{\cos(5x)}.$$

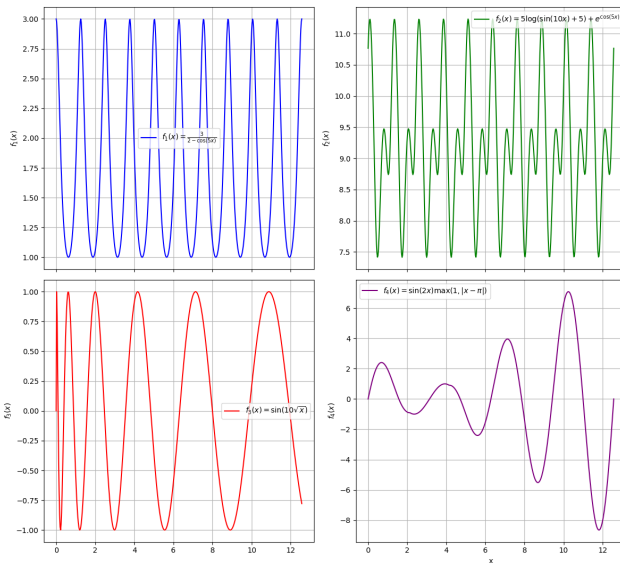
$$f_3(x) = \sin(10\sqrt{x})$$

$$f_4(x) = \sin(2x) \max(1, |x - \pi|).$$

- Compare with spectral-based GLS method.

Synthetic Data

Plots of Four Functions



Simulation Results

Table 1. Comparison for periodic functions (1) $f_1(x) = \frac{3}{2 - \cos(5x)}$ (2) $f_2(x) = 5 \log(\sin(10x) + 5) + e^{\cos(5x)}$. Each entry is the number of detections out of 100 simulations.

err_type	scale	f_1 TDS	f_1 GLS	f_2 TDS	f_2 GLS
–	0.00	85	100	86	100
AG	0.05	98	100	90	100
AL	0.05	100	100	92	100
MG	0.05	100	100	0	100
ML	0.05	99	100	0	100
AG	0.10	100	100	96	100
AL	0.10	100	100	98	100
MG	0.10	15	100	0	100
ML	0.10	10	100	0	100
AG	0.15	94	100	97	100
AL	0.15	62	100	99	100
MG	0.15	0	100	0	100
ML	0.15	0	100	0	100
AG	0.20	3	100	82	100
AL	0.20	0	100	54	100
MG	0.20	0	100	0	100
ML	0.20	0	100	0	100
AG	0.25	0	100	5	100
AL	0.25	0	100	1	100
MG	0.25	0	100	0	99
ML	0.25	0	100	0	100
AG	0.30	0	100	0	100
AL	0.30	0	100	0	100
MG	0.30	0	100	0	93
ML	0.30	0	100	0	94

Table 2. Comparison for non-periodic functions (3) $f_3(x) = \sin(10\sqrt{x})$ (4) $f_4(x) = \sin(2x) \max(1, |x - \pi|)$. Each entry is the number of detections out of 100 simulations.

err_type	scale	f_3 TDS	f_3 GLS	f_4 TDS	f_4 GLS
–	0.00	0	100	11	100
AG	0.05	0	100	8	100
AL	0.05	0	100	5	100
MG	0.05	0	100	1	100
ML	0.05	0	100	3	100
AG	0.10	0	100	1	100
AL	0.10	0	100	0	100
MG	0.10	0	100	0	100
ML	0.10	0	100	0	100

Discussions and Real Data

- Interpretation of the table
 - ▶ As the error scale increased, it increasingly categorized data with multiplicative error as non-periodic
 - ▶ our proposed method is more conservative in terms of classifying a time-series data to be derived from a periodic sampling function.
- For the BIDMC dataset, with arbitrarily chosen 10 fixed values of time delay, our method found among the 53 periodic datasets, 49 datasets were found to be periodic.